

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2325

Anderson Webster Street 100 kV
June 30, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2325. The Interconnection Customer proposes a 74.7 MW solar photovoltaic facility interconnecting at the Anderson Webster Street 100 kV substation in the Indiana area. The request was received on June 23, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 640 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,750,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Rmec to Christopher 100 KV
Contingency: South Haven to Napavine 100 KV line
Loading: 149.5% (Rating: 144 MVA)
DFAX: 57.5%
Pre-project loading: 119.6%
2. Facility: Napavine to South Haven 100 KV
Contingency: Rmec to Christopher 100 KV
Loading: 110.2% (Rating: 185 MVA)
DFAX: 49.2%
Pre-project loading: 90.3%
3. Facility: Wolfe to Barrett 500 KV line
Contingency: Trego to Beaver 500 KV
Loading: 101.1% (Rating: 3,714 MVA)
DFAX: 14.1%
Pre-project loading: 100.8%
4. Facility: Gavin Power Llc to Sporn 230 KV
Contingency: Dillard-Orange to Gavin Power Llc 345 KV L
Loading: 100.2% (Rating: 1,414 MVA)
DFAX: 5.1%
Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rmec to Christopher 100 KV	Rebuild line with higher-capacity conductor	\$3,571,000	52% (\$1,861,000)
South Haven to Napavine 100 KV	Rebuild line with higher-capacity conductor	\$6,784,000	48% (\$3,287,000)
Wolfe to Barrett 500 KV line	Reconductor line	\$115,390,000	0% (\$563,000)
Sporn to Gavin Power Llc 230 KV L	Reconductor line	\$3,722,000	1% (\$39,000)
	Total allocated to project		\$5,750,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.